

10(a)	<i>max 6 of:</i> 1 dendrites (lead to cell body) ; R at both ends 2 nucleus in, cell body / soma ; R if cell body not at one end 3 many mitochondria (in cell body) ; 4 much RER / Nissl's granules (in cell body) ; 5 long / one, <u>axon</u> ; A an <u>axon</u> 6 synaptic, knobs / termini / boutons, at end furthest from cell body ; 7 Schwann cells / myelin ; 8 nodes of Ranvier ; <i>accept points on labelled diagram</i>	6
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Question	Answer	Marks
10(b)	<i>max 9 of:</i> 1 Na^+ / sodium ion, channels open ; I ligand or voltage gated 2 Na^+ enters, cell / axon ; A Na ions / sodium ions 3 inside / p.d., becomes, less negative / positive / +40 mV or causes depolarisation (in correct context) ; 4 Na^+ / sodium ion, channels close ; ecf from mp1 I ligand or voltage-gated 5 K^+ / potassium ion, channels open ; ecf from mp1 I ligand or voltage-gated 6 K^+ moves out (of cell) ; A K ions / potassium ions 7 inside / p.d., becomes negative / A negative figure or causes repolarisation (in correct context) ; 8 <u>local circuits</u> ; 9 myelin (sheath) / Schwann cells, insulate / prevent ion movement ; 10 action potential / depolarisation, <u>only</u> at, nodes (of Ranvier) ; 11 saltatory conduction / action potential jumps from node to node ; A impulse for AP 12 one-way / unidirectional, transmission ; 13 AVP ; e.g. hyperpolarisation / refractory period	9